

LIST OF EXPERIMENTS

1. **The Milgram Experiment:** This experiment was conducted by Stanley Milgram in 1961 to study obedience to authority. It involved participants delivering electric shocks to a “learner” when they answered questions incorrectly.
2. **The Stanford Prison Experiment:** This experiment was conducted by Philip Zimbardo in 1971 to study the psychological effects of becoming a prisoner or prison guard. It involved college students being randomly assigned to play the role of either a prisoner or guard in a simulated prison environment.
3. **The Asch Conformity Experiment:** This experiment was conducted by Solomon Asch in 1951 to study the effects of group pressure on individual behavior. It involved participants being asked to match the length of a line with comparison lines, while in a group where the majority gave incorrect answers.
4. **The Pavlov’s Dog Experiment:** This experiment was conducted by Ivan Pavlov in the 1890s to study classical conditioning. It involved training a dog to associate the sound of a bell with food, causing the dog to salivate at the sound of the bell even when no food was present.
5. **The Bobo Doll Experiment:** This experiment was conducted by Albert Bandura in 1961 to study the effects of observing aggressive behavior. It involved children observing an adult model behaving aggressively towards a Bobo doll, and then being given the opportunity to play with the doll themselves.

Experiments

Note: A minimum of ten experiments from the following should be performed.

1. Experiment 1
2. Experiment 2
3. Experiment 3
4. Experiment 4
5. Experiment 5
6. Experiment 6
7. Experiment 7
8. Experiment 8
9. Experiment 9
10. Experiment 10
11. Experiment 11
12. Experiment 12
13. Experiment 13

14. Experiment 14
15. Experiment 15

(A) Hardware based experiments

Hardware-based experiments are scientific experiments that involve the use of physical equipment or devices to collect data and test hypotheses. These experiments can be conducted in a variety of fields, including physics, chemistry, biology, and engineering.

Some examples of hardware-based experiments include:

1. Measuring the acceleration due to gravity using a simple pendulum.
2. Investigating the properties of materials by conducting tensile strength tests.
3. Studying the behavior of fluids by observing the flow of liquids through pipes or channels.
4. Examining the effects of temperature on the rate of chemical reactions by conducting experiments in a temperature-controlled environment.

Hardware-based experiments can provide valuable insights into the natural world and can help scientists develop new theories and technologies. However, these experiments can be complex and may require specialized equipment and expertise to conduct. Additionally, the results of hardware-based experiments may be influenced by a variety of factors, including the quality of the equipment used and the skill of the experimenter. As such, it is important to carefully design and execute hardware-based experiments to ensure that the results are accurate and reliable.

1. Verification of Kirchhoff's laws

Kirchhoff's laws are two laws that deal with the conservation of charge and energy in electrical circuits. These laws are commonly used to analyze complex circuits and can be verified through experimentation.

- **Kirchhoff's Current Law (KCL):** This law states that the algebraic sum of currents entering and leaving a node in a circuit is equal to zero. In other words, the total current entering a node must be equal to the total current leaving the node.
- **Kirchhoff's Voltage Law (KVL):** This law states that the algebraic sum of the voltage drops around any closed loop in a circuit is equal to zero. In other words, the total voltage supplied to the circuit must be equal to the total voltage used by the circuit.

To verify these laws, an experiment can be conducted using a simple circuit consisting of a power source, resistors, and a multimeter. The multimeter can be used to measure the current at different points in the circuit and the voltage across different components. By comparing the measured values with the

predictions made using Kirchhoff's laws, the validity of the laws can be verified.

2. Measurement of power and power factor in a single phase ac series inductive circuit and study improvement of power factor using capacitor

- Power is the rate at which energy is transferred or used. In an AC circuit, power can be calculated by multiplying the voltage and current, and then multiplying by the power factor.
- The power factor is the ratio of the real power to the apparent power. It is a measure of how effectively the current is being converted into useful work output.
- In a single phase AC series inductive circuit, the power factor is less than 1 due to the presence of inductance. This means that the circuit is not using all of the power that is being supplied to it.
- The power factor can be improved by adding a capacitor in parallel with the inductive load. This is known as power factor correction.
- The capacitor provides a leading reactive power that cancels out the lagging reactive power of the inductive load, thus improving the power factor.
- By improving the power factor, the circuit can use more of the power that is being supplied to it, resulting in increased efficiency and reduced energy costs.

3. Study of phenomenon of resonance in RLC series circuit and obtain resonant frequency.

Resonance in an RLC series circuit occurs when the inductive and capacitive reactances are equal in magnitude, causing the circuit to exhibit a purely resistive impedance. This results in maximum current flow through the circuit.

The resonant frequency of an RLC series circuit can be calculated using the formula:

$$f = 1 / (2 * \sqrt{LC})$$

Where: - f is the resonant frequency in hertz (Hz) - L is the inductance in henries (H) - C is the capacitance in farads (F)

At the resonant frequency, the impedance of the circuit is equal to the resistance, and the voltage and current are in phase. The circuit is said to be in resonance.

The phenomenon of resonance can be observed by varying the frequency of the applied voltage and measuring the resulting current through the circuit. At the resonant frequency, the current will reach a maximum value.

In summary, resonance in an RLC series circuit occurs when the inductive and capacitive reactances are equal, resulting in maximum current flow through the circuit. The resonant frequency can be calculated using the formula $f = 1 / (2 * \sqrt{LC})$. At the resonant frequency, the circuit exhibits a purely resistive

impedance and the voltage and current are in phase. This phenomenon can be observed by varying the frequency of the applied voltage and measuring the resulting current through the circuit.

4. Connection and measurement of power consumption of a fluorescent lamp (tube light)

A fluorescent lamp, also known as a tube light, is a low-pressure mercury-vapor gas-discharge lamp that uses fluorescence to produce visible light. To measure the power consumption of a fluorescent lamp, the following steps can be taken:

1. **Connect the lamp to a power source:** The fluorescent lamp should be connected to a power source using the appropriate wiring and connections. Ensure that the connections are secure and that the lamp is properly installed.
2. **Measure the current:** Use a clamp meter to measure the current flowing through the circuit. The clamp meter should be placed around one of the wires leading to the lamp.
3. **Measure the voltage:** Use a multimeter to measure the voltage across the lamp. The multimeter should be connected to the two wires leading to the lamp.
4. **Calculate the power consumption:** The power consumption of the lamp can be calculated by multiplying the current and voltage measurements. The result will be the power consumption in watts.

It is important to note that the power consumption of a fluorescent lamp may vary depending on factors such as the age of the lamp, the temperature of the environment, and the quality of the power supply. It is recommended to take multiple measurements to obtain an accurate average power consumption.

5. Measurement of power in 3- phase circuit by two-wattmeter method and determination of its power factor for star as well as delta connected load.

The two-wattmeter method is a common technique used to measure power in a three-phase circuit. This method involves the use of two wattmeters, which are connected to measure the power in two of the three phases. The third phase is calculated by subtracting the sum of the first two phases from the total power.

The power factor of the circuit can be determined by measuring the phase angle between the voltage and current in each phase. This can be done using a power factor meter or by calculating the power factor using the readings from the two wattmeters.

In a star-connected load, the power factor can be calculated by dividing the total power by the product of the line voltage and the line current. In a delta-

connected load, the power factor can be calculated by dividing the total power by the product of the phase voltage and the phase current.

The two-wattmeter method is a simple and accurate way to measure power in a three-phase circuit and determine its power factor. It is commonly used in industrial and commercial settings to monitor and control the power consumption of three-phase equipment.

6. Determination of parameters of ac single phase series RLC circuit

An RLC circuit is an electrical circuit consisting of a resistor (R), an inductor (L), and a capacitor (C), connected in series or in parallel. The name of the circuit is derived from the letters that are used to denote the constituent components of this circuit, where the sequence of the components may vary from RLC.

The circuit forms a harmonic oscillator for current and resonates similarly to an LC circuit. The main difference that the presence of the resistor makes is that any oscillation induced in the circuit will die away over time if it is not kept going by a source. This effect of the resistor is called damping. The presence of the resistor also reduces the peak resonant frequency somewhat.

The behavior of the circuit is described by the following differential equation:

$$\frac{d^2q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{1}{LC}q = \frac{V(t)}{L}$$

where q is the electric charge on the capacitor, $V(t)$ is the voltage source, and R , L , and C are the resistance, inductance, and capacitance, respectively.

The solution to this equation is a damped sinusoid, which describes the oscillation of the charge on the capacitor. The frequency of the oscillation is determined by the values of L and C , while the rate at which the oscillation dies away is determined by the value of R .

The parameters of an AC single-phase series RLC circuit can be determined by measuring the voltage and current at different points in the circuit and using Ohm's law and Kirchhoff's laws to calculate the values of R , L , and C . Alternatively, the values of R , L , and C can be determined by measuring the resonant frequency of the circuit and the rate at which the oscillation dies away. These measurements can be made using an oscilloscope or a frequency analyzer.

7. Determination of (i) Voltage ratio (ii) polarity and (iii) efficiency by load test of a single phase Transformer

A single-phase transformer is an electrical device that is used to transfer electrical energy from one circuit to another by means of electromagnetic induction. The voltage ratio, polarity, and efficiency of a single-phase transformer can be determined by performing a load test.

- i. **Voltage Ratio:** The voltage ratio of a transformer is the ratio of the secondary voltage to the primary voltage. It can be determined by measuring the primary and secondary voltages of the transformer while it is under load.
- ii. **Polarity:** The polarity of a transformer refers to the direction of the induced voltage in the secondary winding relative to the primary winding. It can be determined by performing a polarity test, which involves connecting the primary and secondary windings in series and measuring the resulting voltage.
- iii. **Efficiency:** The efficiency of a transformer is the ratio of the output power to the input power. It can be determined by measuring the input and output power of the transformer while it is under load.

In summary, the voltage ratio, polarity, and efficiency of a single-phase transformer can be determined by performing a load test. This involves measuring the primary and secondary voltages, performing a polarity test, and measuring the input and output power of the transformer while it is under load. These measurements can provide valuable information about the performance of the transformer and can help to ensure that it is operating efficiently and effectively.

8. Determination of efficiency of a dc shunt motor by load test

The efficiency of a DC shunt motor can be determined by performing a load test. The load test involves measuring the input power, output power, and losses of the motor at different load conditions. The following steps can be followed to determine the efficiency of a DC shunt motor by load test:

1. Connect the DC shunt motor to a suitable load and power supply.
2. Measure the input current and voltage to the motor using an ammeter and voltmeter, respectively.
3. Measure the speed of the motor using a tachometer.
4. Calculate the input power to the motor by multiplying the input current and voltage.
5. Calculate the output power of the motor by multiplying the torque and speed of the motor.
6. Calculate the losses in the motor by subtracting the output power from the input power.
7. Calculate the efficiency of the motor by dividing the output power by the input power and multiplying by 100%.

It is important to note that the efficiency of a DC shunt motor will vary with the load. Therefore, it is necessary to perform the load test at different load conditions to obtain an accurate measure of the motor's efficiency. Additionally, the accuracy of the measurements and calculations will affect the accuracy of the efficiency determination. It is important to use accurate instruments and perform the calculations carefully to obtain reliable results.

9. To study running and speed reversal of a three phase induction motor and record speed in both directions.

A three-phase induction motor is an AC motor that operates on the principle of electromagnetic induction. The motor consists of a stator, which is a stationary part, and a rotor, which is a rotating part. The stator has a set of windings that are energized by a three-phase AC power supply. This creates a rotating magnetic field that induces a current in the rotor windings, causing the rotor to rotate.

The speed of a three-phase induction motor can be controlled by varying the frequency of the AC power supply. The speed of the motor is directly proportional to the frequency of the power supply. Therefore, by increasing or decreasing the frequency of the power supply, the speed of the motor can be increased or decreased.

The direction of rotation of a three-phase induction motor can be reversed by changing the sequence of the power supply to the stator windings. This can be achieved by interchanging any two of the three-phase power supply connections to the motor. When the sequence of the power supply is changed, the direction of the rotating magnetic field is reversed, causing the rotor to rotate in the opposite direction.

To study the running and speed reversal of a three-phase induction motor, the following steps can be followed:

1. Connect the three-phase induction motor to a variable frequency drive (VFD) that can vary the frequency of the power supply to the motor.
2. Start the motor and observe its speed and direction of rotation.
3. Vary the frequency of the power supply using the VFD and observe the change in speed of the motor.
4. Record the speed of the motor at different frequencies.
5. Stop the motor and change the sequence of the power supply connections to the motor to reverse its direction of rotation.
6. Start the motor and observe its speed and direction of rotation.
7. Vary the frequency of the power supply using the VFD and observe the change in speed of the motor.
8. Record the speed of the motor at different frequencies in the reversed direction.

By following these steps, the running and speed reversal of a three-phase induction motor can be studied and the speed of the motor can be recorded in both directions.

10. Demonstration of cut-out sections of machines: dc machine, three phase induction machine, single-phase induction machine and synchronous machine.

A cut-out section of a machine is a physical model that shows the internal

components and workings of the machine. It is created by removing a section of the machine's casing to reveal the internal components. Cut-out sections are commonly used for educational and demonstration purposes.

1. **DC Machine:** A DC machine is an electrical machine that converts mechanical energy into direct current (DC) electrical energy or vice versa. A cut-out section of a DC machine would typically show the armature, commutator, brushes, and field windings.
2. **Three Phase Induction Machine:** A three phase induction machine is an AC electric motor that converts electrical energy into mechanical energy. A cut-out section of a three phase induction machine would typically show the stator, rotor, and windings.
3. **Single-Phase Induction Machine:** A single-phase induction machine is an AC electric motor that operates on single-phase power. A cut-out section of a single-phase induction machine would typically show the stator, rotor, and windings.
4. **Synchronous Machine:** A synchronous machine is an AC electric machine that operates at a constant speed, determined by the frequency of the power source. A cut-out section of a synchronous machine would typically show the rotor, stator, and field windings.

These cut-out sections provide a visual representation of the internal workings of the machines, allowing for a better understanding of their operation and construction. They are commonly used in educational settings to demonstrate the principles of electromechanical energy conversion.

(B) Experiments available on virtual lab

Virtual labs provide a platform for students to conduct experiments and simulations online. There are a variety of virtual labs available that offer a wide range of experiments and simulations. Some of the experiments available on virtual labs are:

1. **Virtual Chemistry Experiments:** A collection of interactive web-based chemistry tutorials .
2. **Mixed Reception:** Solve a mysterious death by gathering clues, interviewing suspects, and analyzing evidence .
3. **The Virtual Lab:** An online simulation of a chemistry lab designed to help students link chemical computations with authentic laboratory chemistry .
4. **Virtual Experiments:** Adaptive to any learning style, with progress tracking, walkthrough videos, lab manuals, enrichment documents, immediate feedback, MCQ quizzes, and much more .
5. **VRLab Academy:** More than 200 science experiments available on VR and PC, with virtual reality technology to increase the ability to learn through virtual memory .

6. **Aspirin Virtual Chemistry:** Explore the chemistry of aspirin virtually with four levels of experiments, including synthesis, thin layer chromatography, and reaction conditions .
7. **CK-12 Chemistry Simulations:** Nearly two dozen simulations cover topics like average atomic mass, solubility with rock candy, and freezing point depression with road salt .
8. **Mystery Science:** Offers a wealth of virtual science labs, activities, and experiments for students in elementary school, covering topics from weather to astronomy, humans to animals .
9. **Molecular Workbench:** Offers hundreds of virtual science labs for physics, biology, chemistry, and more .
10. **National Science Digital Library:** ChemCollective Virtual Labs, take a deep dive into chemistry with these virtual labs .
11. **PBS Nova Labs:** A terrific platform for teens and older learners to explore and examine cutting-edge issues .

These are just a few examples of the many experiments available on virtual labs. Virtual labs provide a convenient and accessible way for students to conduct experiments and simulations, and to learn about science in an interactive and engaging way.

1. Kirchhoff's laws

Kirchhoff's laws are two laws that deal with the conservation of charge and energy in electrical circuits. These laws are named after the German physicist Gustav Kirchhoff.

1. Kirchhoff's Current Law (KCL): This law states that the algebraic sum of currents entering a node (or a closed boundary) is zero. In other words, the total current entering a node must be equal to the total current leaving the node. This law is based on the principle of conservation of electric charge.
2. Kirchhoff's Voltage Law (KVL): This law states that the algebraic sum of the potential differences in any closed loop must be zero. In other words, the total voltage around any closed loop in a circuit must be zero. This law is based on the principle of conservation of energy.

These laws are widely used in circuit analysis to determine the current, voltage, and power in a circuit. They are fundamental principles that must be satisfied in any circuit, regardless of its complexity.

Virtual Lab

Virtual Lab is an initiative of the Ministry of Human Resource Department under the National Mission on Education through ICT. The project aims to provide remote access to labs in various disciplines of Science and Engineering

for students at the undergraduate level, postgraduate level, and research scholars. The main objective of the Virtual Labs is to enthuse students to conduct experiments by arousing their curiosity, which would help them in learning basic and advanced concepts through remote experimentation .

The Virtual Lab website provides a range of experiments and labs that are hosted for open access. One such experiment is the Kirchhoff's Laws experiment, which can be accessed through the simulator link provided on the website. The experiment involves the use of a resistor, which is a two-terminal electronic component that produces a voltage across its terminals that is proportional to the electric current flowing through it in accordance with Ohm's law . The experiment is based on Kirchhoff's Voltage Law, which states that the algebraic sum of all voltages in a loop must equal zero .

2. Thevenin Theorem

Thevenin's Theorem is a method used to simplify complex electrical circuits. It states that any linear, bilateral circuit can be replaced by an equivalent circuit consisting of a single voltage source and a single resistor.

The steps to apply Thevenin's Theorem are as follows: 1. Identify the load resistor in the circuit. 2. Remove the load resistor from the circuit. 3. Calculate the open-circuit voltage across the terminals where the load resistor was connected. This is the Thevenin voltage. 4. Calculate the equivalent resistance of the circuit as seen from the open terminals. This is the Thevenin resistance. 5. Replace the original circuit with the Thevenin equivalent circuit, consisting of the Thevenin voltage source in series with the Thevenin resistance. 6. Reconnect the load resistor to the Thevenin equivalent circuit.

Thevenin's Theorem can be used to analyze circuits with multiple sources and multiple resistors. It is a powerful tool for simplifying complex circuits and making them easier to analyze.

Virtual Lab

- Virtual Labs is a project initiated by the Ministry of Education, Government of India, under the National Mission on Education through Information and Communication Technology.
- The project aims to provide remote access to Laboratories in various disciplines of Science and Engineering for students at all levels from undergraduate to research.

3. RLC Series Resonance

An RLC series circuit consists of a resistor (R), an inductor (L), and a capacitor (C) connected in series. At a certain frequency, called the resonant frequency, the impedance of the circuit is at its minimum value. This is known as series resonance.

- The resonant frequency of an RLC series circuit is given by the formula: $f = 1 / (2 * \pi * \sqrt{L * C})$.
- At resonance, the inductive reactance and capacitive reactance are equal in magnitude and opposite in sign, so they cancel each other out.
- The total impedance of the circuit at resonance is equal to the resistance of the resistor.
- The current in the circuit is at its maximum value at resonance.
- The voltage across the inductor and capacitor are equal in magnitude and opposite in sign at resonance.
- The power factor of the circuit is 1 at resonance, meaning that all the power supplied by the source is dissipated in the resistor.
- The bandwidth of the circuit, which is the range of frequencies over which the circuit responds, is determined by the resistance and the quality factor (Q) of the circuit.
- The quality factor is given by the formula: $Q = (2 * \pi * f * L) / R$, where f is the resonant frequency.
- A higher quality factor means a narrower bandwidth and a sharper resonance.

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4. Measurement of power in 3- phase circuit by two wattmeter method and determination of its power factor for star as well as delta connected load.

The two wattmeter method is a common method for measuring power in a three-phase circuit. This method involves the use of two wattmeters to measure the power consumed by a three-phase load. The power factor of the load can also be determined using this method.

The two wattmeter method can be used for both star and delta connected loads. The procedure for measuring power and determining the power factor is slightly different for each type of connection.

For a star connected load, the two wattmeters are connected between the line and the neutral. The readings of the two wattmeters are added to obtain the total power consumed by the load. The power factor can be determined by dividing the total power by the product of the line voltage and the line current.

For a delta connected load, the two wattmeters are connected between the lines. The readings of the two wattmeters are added to obtain the total power

consumed by the load. The power factor can be determined by dividing the total power by the product of the line voltage and the line current.

In summary, the two wattmeter method is a useful and accurate method for measuring power in a three-phase circuit and determining the power factor of the load. This method can be used for both star and delta connected loads. The procedure for measuring power and determining the power factor is slightly different for each type of connection.

Virtual Lab

Virtual Lab is an online platform that allows students to perform experiments and learn about science and engineering subjects. The experiments are designed either as simulations or as remote triggered. A remote triggered lab allows a user to connect to real equipment using a web browser.

One such virtual lab is the Virtual Power Laboratory-D.E.I. This lab is designed to facilitate students to work with their own convenience and easily understand the practical concepts. It also helps in generating interest in the student community to get better insight about the subjects related to machines and power systems.

The Virtual Power Laboratory-D.E.I offers various experiments, including the study of the synchronization of an alternator with an infinite bus bar, and the determination of the direct axis reactance (X_d) and quadrature axis reactance (X_q) of a synchronous machine.

In the measurement section of the Virtual Power Laboratory-D.E.I, students can learn about various measuring devices, such as ammeters, voltmeters, wattmeters, and power factor meters. They can also learn about different techniques for power measurement.

5. Determination of parameters of ac single phase series RLC circuit

An RLC circuit is an electrical circuit consisting of a resistor (R), an inductor (L), and a capacitor (C), connected in series. The behavior of an RLC circuit is determined by the values of the resistance, inductance, and capacitance, as well as the frequency of the applied alternating current (ac) voltage.

To determine the parameters of an ac single phase series RLC circuit, the following steps can be followed:

1. Measure the resistance of the resistor using an ohmmeter.
2. Measure the inductance of the inductor using an inductance meter.
3. Measure the capacitance of the capacitor using a capacitance meter.
4. Calculate the impedance of the circuit using the formula $Z = R + j(X_L - X_C)$, where X_L is the inductive reactance, X_C is the capacitive reactance,

and j is the imaginary unit.

5. Calculate the resonant frequency of the circuit using the formula $f = 1 / (2 * \pi * \sqrt{L * C})$.
6. Calculate the quality factor of the circuit using the formula $Q = (2 * \pi * f * L) / R$.
7. Calculate the bandwidth of the circuit using the formula $BW = f / Q$.

These are the basic steps for determining the parameters of an ac single phase series RLC circuit. It is important to note that the accuracy of the measurements and calculations will depend on the precision of the instruments used and the skill of the person performing the measurements and calculations.

Virtual Lab

Virtual Lab is an initiative by Amrita University, Dayalbagh University, NIT Karnataka, and COE Pune. The main objective of Virtual Labs is to provide remote access to labs in various disciplines of Science and Engineering. These Virtual Labs cater to students at the undergraduate level, postgraduate level, and research scholars .

Amrita Virtual Biotechnology laboratory is a new kind of experimental science that is being established as a virtual simulation-based laboratory. These labs focus on helping students retain the real feel of a laboratory while conducting the experiment from an internet-enabled computer terminal, much in the same way as is done in a real lab .

Amrita Vishwa Vidyapeetham is transforming the face of higher education and research through a revolutionary new educational tool – virtual labs. The VALUE Initiative is a project dedicated to developing these virtual labs in various science and engineering disciplines .

Virtual Lab at Amrita uses state-of-the-art computer simulation technology to create real-world environments and problem-handling capabilities. It is required to bridge the gap between institutions (or industries) that retain the physical laboratory and distantly placed economically challenged educational institutions in areas such as Physics .

The Online Labs is based on the idea that lab experiments can be taught using the Internet, more efficiently and less expensively. The labs can also be made available to students with no access to physical labs or where equipment is not available owing to being scarce or costly .

6. To observe the B-H loop of a ferromagnetic material in CRO

A B-H loop, also known as a hysteresis loop, is a graphical representation of the relationship between the magnetic field strength (H) and the magnetic flux density (B) of a ferromagnetic material. The loop is obtained by subjecting the material to an alternating magnetic field and measuring the resulting magnetization.

To observe the B-H loop of a ferromagnetic material in a Cathode Ray Oscilloscope (CRO), the following steps can be followed:

1. Prepare a sample of the ferromagnetic material in the form of a toroid or a solenoid.
2. Connect the sample to a primary coil and a secondary coil.
3. Connect the primary coil to an alternating current (AC) power supply.
4. Connect the secondary coil to the vertical input of the CRO.
5. Connect a search coil to the horizontal input of the CRO.
6. Adjust the CRO settings to display the B-H loop on the screen.
7. Observe the B-H loop on the CRO screen.

The shape of the B-H loop provides information about the magnetic properties of the ferromagnetic material, such as its coercivity, remanence, and saturation magnetization. These properties can be used to determine the suitability of the material for various applications, such as in the design of electromagnets and transformers.

Virtual Lab

Virtual labs are online tools that offer a space for students to engage with their subject matter interactively. They can be used for pre-lab preparation, online lab courses, post-lab review, make-up labs, or used with a textbook to bridge the gap between lab and lecture courses. Virtual labs are available for a variety of disciplines, including Anatomy & Physiology, Biology (Majors and Non-majors), Botany, Environmental Science, Genetics, Human Biology, Medical Assisting, Microbiology, and Nutrition.

Virtual labs can be used to provide remote-access to simulation-based labs in various disciplines of Science and Engineering. They can help students learn basic and advanced concepts through remote experimentation. Virtual labs can also include built-in quizzes, notebooks, and references to help students learn and retain information.

Virtual labs are a valuable resource for students and educators, as they provide an interactive and engaging way to learn and teach complex concepts. They can be used in a variety of settings, including remote learning, and can help bridge the gap between theory and practice.

7. Determination of the efficiency of a dc motor by loss summation method (Swinburne's test)

Swinburne's test is an indirect method of testing DC machines, named after Sir James Swinburne. It is the simplest method of testing shunt and compound DC machines which have constant field flux. In this method, the DC machine (generator or motor) is run as a motor at no-load and losses of the machine are determined. Once the losses of the machine are known, its efficiency at any desired load can be determined in advance.

The losses in the motors are armature copper losses, iron losses in the core, friction losses, and winding losses. These losses are calculated separately and efficiency can be pre-determined .

Course Outcomes:

- Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course.
- They are used to communicate the goals and expectations of the course to students, and to assess the effectiveness of the course in achieving those goals.
- Course outcomes should be specific, measurable, and achievable within the scope of the course.
- They should be aligned with the overall program outcomes and the institution's mission and goals.
- Course outcomes should be regularly reviewed and updated to ensure that they remain relevant and reflect current best practices in the field.
- Effective course outcomes can help students understand the purpose and value of the course, and can guide their learning and self-assessment.
- They can also help instructors design and deliver effective instruction, and can provide a basis for evaluating and improving the course.

Course Outcome (CO) Bloom's Level

Course Outcome (CO) refers to the expected knowledge, skills, and abilities that a student should possess upon completion of a specific course. These outcomes are used to assess the effectiveness of the course and to ensure that the course content aligns with the overall program goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. The levels of Bloom's Taxonomy, in ascending order of complexity, are: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating.

The Bloom's Level of a Course Outcome refers to the level of complexity and specificity of the outcome, as defined by Bloom's Taxonomy. For example, a Course Outcome that requires students to recall factual information would be at the Remembering level of Bloom's Taxonomy, while an outcome that requires students to apply knowledge to solve problems would be at the Applying level.

In summary, the Course Outcome (CO) Bloom's Level is a measure of the complexity and specificity of the expected knowledge, skills, and abilities that a student should possess upon completion of a specific course, as defined by Bloom's Taxonomy. It is used to assess the effectiveness of the course and to ensure alignment with the overall program goals.

At the end of this course, the students should be able to:

1. Understand the fundamental concepts and principles of the subject.
2. Apply the knowledge and skills acquired in the course to solve problems and make informed decisions.
3. Communicate effectively using the language and terminology of the subject.
4. Work collaboratively with others to achieve common goals.
5. Demonstrate critical thinking and analytical skills.
6. Develop a lifelong interest in learning and self-improvement.

CO 1 Conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits. K3

- KVL (Kirchhoff's Voltage Law) states that the sum of the voltage drops around a closed loop in a circuit is equal to the sum of the voltage sources in that loop.
- KCL (Kirchhoff's Current Law) states that the sum of the currents entering a node in a circuit is equal to the sum of the currents leaving that node.
- Network theorems, such as Thevenin's theorem and Norton's theorem, can be used to simplify complex circuits and make it easier to analyze them.
- To conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits, you can set up a simple circuit with a power source, resistors, and measuring devices such as a voltmeter and an ammeter.
- You can then use KVL to verify that the sum of the voltage drops around the circuit is equal to the voltage of the power source.
- You can also use KCL to verify that the sum of the currents entering and leaving a node in the circuit is equal.
- You can also apply network theorems to simplify the circuit and make it easier to analyze.
- These experiments can help you understand the fundamental principles of circuit analysis and the application of KVL/KCL and network theorems to DC electrical circuits.

CO 2 Demonstrate the behavior of AC circuits connected to single phase AC supply and measure power in single phase as well as three phase electrical circuits. K4

- AC circuits are electrical circuits that are powered by an alternating current (AC) source.
- A single-phase AC supply is an electrical power supply that delivers power through a single conductor.
- The behavior of an AC circuit connected to a single-phase AC supply can be demonstrated by analyzing the voltage and current waveforms in the circuit.
- Power in a single-phase AC circuit can be measured using a power meter

or by calculating the product of the RMS voltage and current values.

- A three-phase electrical circuit is a type of polyphase system that uses three conductors to deliver power.
- Power in a three-phase electrical circuit can be measured using a three-phase power meter or by calculating the sum of the power in each phase.
- It is important to understand the behavior of AC circuits and to be able to measure power in single-phase and three-phase electrical circuits in order to design and operate electrical systems effectively.

CO 3 Perform experiment illustrating BH curve of magnetic materials. K3

The BH curve, also known as the hysteresis loop, is a graphical representation of the relationship between the magnetic field strength (H) and the magnetic flux density (B) of a magnetic material. It is used to characterize the magnetic properties of a material.

To perform an experiment illustrating the BH curve of magnetic materials, the following steps can be followed:

1. Prepare a sample of the magnetic material to be tested.
2. Place the sample in a magnetic field and measure the magnetic flux density (B) as the magnetic field strength (H) is increased.
3. Plot the measured values of B against H to obtain the initial magnetization curve.
4. Reduce the magnetic field strength to zero and measure the residual magnetism of the sample.
5. Reverse the direction of the magnetic field and increase its strength, measuring the magnetic flux density at each step.
6. Plot the measured values of B against H to obtain the hysteresis loop.
7. Repeat the process of increasing and decreasing the magnetic field strength to obtain a complete BH curve.

The shape of the BH curve can provide information about the magnetic properties of the material, such as its coercivity, remanence, and saturation magnetization. These properties can be used to classify the material as hard or soft magnetic material and to determine its suitability for various applications.

CO 4 Calculate efficiency of a single phase transformer and DC machine. K4

Efficiency is the ratio of output power to input power. It is a measure of how well a device converts the power it receives into useful work. The efficiency of a single phase transformer and a DC machine can be calculated using the following steps:

1. **Determine the input power:** Measure the voltage and current at the input of the device and use the formula $P = VI$ to calculate the input

power.

2. **Determine the output power:** Measure the voltage and current at the output of the device and use the formula $P = VI$ to calculate the output power.
3. **Calculate the efficiency:** Divide the output power by the input power and multiply by 100 to express the efficiency as a percentage. The formula for efficiency is $\text{Efficiency} = (\text{Output Power} / \text{Input Power}) * 100$.

For a single phase transformer, the input power is the power supplied to the primary winding and the output power is the power delivered by the secondary winding. For a DC machine, the input power is the electrical power supplied to the armature and the output power is the mechanical power delivered by the shaft.

It is important to note that the efficiency of a device is not constant and can vary depending on the operating conditions. Factors such as load, temperature, and frequency can affect the efficiency of a single phase transformer and a DC machine. Therefore, it is important to measure the efficiency under the specific operating conditions for which the device is intended to be used.

CO 5 Perform experiments on speed measurement and reversal of direction of three phase induction motor and Identify the type of DC and AC machines based on their construction. K4

1. To perform experiments on speed measurement and reversal of direction of three phase induction motor, the following steps can be followed:
 - Connect the three phase induction motor to a power source and a speed measurement device.
 - Measure the speed of the motor using the speed measurement device.
 - To reverse the direction of the motor, interchange any two of the three power supply lines.
 - Measure the speed of the motor again to confirm that the direction has been reversed.
2. To identify the type of DC and AC machines based on their construction, the following characteristics can be observed:
 - DC machines have a commutator and brushes, while AC machines do not.
 - The armature windings of a DC machine are on the rotor, while the armature windings of an AC machine are on the stator.
 - The magnetic field of a DC machine is produced by electromagnets, while the magnetic field of an AC machine is produced by permanent magnets or by a rotating magnetic field.

These are some basic steps and characteristics that can be used to perform experiments on speed measurement and reversal of direction of three phase induction motor and to identify the type of DC and AC machines based on

their construction. It is important to follow proper safety procedures while performing these experiments.

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

These are the six levels of Bloom’s Taxonomy, a framework for categorizing educational goals and objectives. The taxonomy is hierarchical, meaning that the higher levels build upon the lower levels.

1. **K1 – Remember:** This level involves recalling information from memory. Examples of tasks at this level include defining terms, listing facts, and identifying concepts.
2. **K2 – Understand:** This level involves comprehending the meaning of information. Examples of tasks at this level include explaining concepts, summarizing information, and interpreting data.
3. **K3 – Apply:** This level involves using information in a new situation. Examples of tasks at this level include solving problems, applying theories, and demonstrating techniques.
4. **K4 – Analyze:** This level involves breaking down information into its component parts and understanding their relationships. Examples of tasks at this level include comparing and contrasting, categorizing, and identifying cause and effect.
5. **K5 – Evaluate:** This level involves making judgments about the value of information. Examples of tasks at this level include critiquing, assessing, and justifying.
6. **K6 – Create:** This level involves combining information to form a new whole. Examples of tasks at this level include designing, constructing, and planning.

These levels can be used to design educational activities and assessments that target different levels of cognitive complexity. They can also be used to evaluate the level of understanding that a student has achieved.